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Roll No

EX-6002-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Power System-II

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Why the interconnected power system is necessary?
Explain the problems associated with them.
b) What is deregulation in power system? Explain its effect
and how it can be overcome.
2. a) What is distributed generation? Explain utility and
customer benefits of using distributed generation.
b) Explain to calculate available transfer capacities using ac
power flow analysis.
3. a) What is load flow solution? Explain its significance in
power system analysis.
b) Classify various types of buses in a power system for load
flow studies. Justify the classification.
4. a) Explain clearly with a flow chart the computational
procedure for load flow solution using Gauss Seidel
method when the system contains all types of buses.

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- b) Incremental fuel costs in Rs. per mega watt hour for two units in a plant are given by

$$\frac{dF_1}{dP_1} = 0.1P_1 + 20, \quad \frac{dF_2}{dP_2} = 0.12P_2 + 16$$

The minimum and maximum loads on each unit are to be 20MW and 125MW respectively. Determine the incremental fuel cost and the allocation of load between units for the minimum cost when loads are

i) 100MW

ii) 150MW

Assume both unit are operating.

5. a) What is single area system? Deduce dynamic models for a single area system. Show the full block diagram.
b) Why load frequency is necessary? Explain types of frequency regulation with proper example.
6. a) Explain with the help of block diagram automatic voltage regulator of turbo generators.
b) Discuss in detail about generation and absorption of reactive power in power system components.
7. a) What is equal area criteria? Explain.
b) Derive swing equation and discuss its application in the study of power system stability.
8. Explain the following.
a) Economic dispatch
b) Synchronous phase modifier
c) Method improving transient stability
d) Static VAR systems

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